

Spec-1: Justifications and Decisions (Compute Platform)

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Iteration history

This spec went through three rounds before reaching the version submitted:

1. **First draft** (FR/NFR style): stakeholders, functional requirements, non-functional requirements, in/out of scope. Not yet in the required 7-section format, and contained a self-contradiction (an operational "dashboard" listed as a requirement while UI was simultaneously listed out of scope), several non-testable requirements ("must not monopolize," "understandable to non-expert students"), and several unspecified interaction rules (queueing, admission control, reservation-vs-preemption behavior).
2. **Second draft** (Contract-heavy iteration): resolved every ambiguity from round one with concrete rules (SIGTERM + 60s grace period, FIFO with TTL, explicit AND condition on dual quota pools, named error codes, entity and endpoint shapes). This introduced a new set of issues: a REST/JSON contract that conflicted with the brief's instruction not to lock the tech stack, invented numeric thresholds with no stated rationale, an unresolved token-quota timing question (tokens are only known after generation, not at admission), no rule for two overlapping instructor reservations, ambiguous wording on hard vs. soft resource unavailability, a `role` field that quietly reintroduced RBAC despite the text claiming that model was undecided, dropped audit logging (present in round one, missing in round two), and unranked assumptions.
3. **Third draft**: every issue from round two resolved, then condensed into the required 7-section format under the page limit. This condensation, done to satisfy the 1–2 page constraint, cut the entity/endpoint precision gained in round two back down to concept-level field lists, which reintroduced a milder version of the original Contract-precision problem.
4. **This version**, after an external review pass: the review (reproduced in full in the submission folder) scored this document 88/100 and flagged three concrete, fixable gaps rather than a rewrite. Applied here: (a) each Contract action now states explicit input and output fields, without naming a transport or serialization format, closing the gap between "conceptual entities" and "exact signatures" the review identified; (b) the token-quota timing decision (reserve an estimated cap at admission, truncate at that cap, reconcile after completion), previously stated only in this justifications document, is now stated directly in the Contract, so the two documents no longer disagree; (c) the previously unstated behavior for a job that does not checkpoint within its grace period is now explicit (forced termination, resources reclaimed, recorded as preempted rather than failed). A fourth, smaller change requested directly by the author: the "equivalent-or-better" hardware comparison, which the review noted was left undefined, now carries an explicit sentence stating that this is a deliberate scope deferral (comparing it would mean designing internal scheduling logic) rather than an oversight, and the same item was added to Non-Goals so the deferral is stated in two places instead of implied in one.

Key decisions and why

Transport framing (Contract). Chosen: keep the request/response fields and error codes as the binding contract, explicitly label any transport detail (REST path, HTTP verbs) as illustrative only. Rejected alternative: commit to REST/JSON now, since the assignment asks for "exact function/API signatures." I chose illustrative framing because the course brief for this specific project explicitly forbids locking the technology stack in Iteration 1, and that instruction outranks the general Spec-1 template when the two conflict. The Contract still satisfies "no implementation leakage" and "testable" because the binding part (field names, types, error taxonomy) does not depend on which transport carries it.

Reservation conflict resolution. Chosen: an escalating sequence, first-come-first-served, then an automatic substitute offer on equivalent-or-better hardware, then admin review only if neither resolves it. Rejected alternatives considered: reject the second request outright (simplest, but defeats the stated goal of minimizing admin bottlenecks and fair self-coordination), or send every conflict straight to admin (defeats the same goal from the other direction, since it makes a human review a bottleneck for something the system could resolve itself most of the time). The chosen option was a direct response to the stated project goal of efficient, low-friction, fair coordination, so it is grounded in the brief's own stated intent, not an invented feature.

Hard vs. soft resource unavailability. Chosen: reject only when the required hardware class does not exist anywhere in the fleet; queue when it exists but is currently busy. Rejected alternative: reject in both cases, which was the original wording and would have caused the system to refuse requests that a human operator would obviously just wait out.

Token quota timing. Chosen: reserve an estimated cap at admission (checked against the quota pools), truncate generation at that cap if exceeded, reconcile the ledger to actual usage afterward. This is a standard pre-emptive metering pattern; the

alternative (check only after generation completes) would let a single job overdraw a shared course pool before anyone could stop it, which conflicts directly with the fairness goal stated by Academic Administration and Teachers in the stakeholder table.

Numeric thresholds. Chosen: mark all specific numbers (grace periods, TTLs) as adjustable policy defaults rather than fixed technical constraints, and moved this framing into Constraints so it applies globally instead of repeating a caveat next to every number. This was a direct fix for the "invented constraints" risk called out in the assignment's evaluation criteria; the numbers stay because Success Signal needs something measurable, but they are no longer presented as authoritative.

`role` field vs. undecided authorization model. Chosen: keep role but scope it explicitly to coarse, structural capability (what categories of action are possible), separate from fine-grained authorization (which specific models/courses a user can reach), which remains undecided. Rejected: removing role entirely, which would have made the Contract unable to express even basic distinctions like "only instructors can create reservations," a capability explicitly required by the brief.

Audit logging. Reinstated as a hard requirement after it was dropped between draft one and draft two. This was not a new decision so much as catching a regression: draft one had it, draft two silently lost it while gaining contract precision elsewhere. It is now stated once, generally (any policy bypass or state-changing administrative action), rather than repeated per capability.

AI usage tracking

Claude was used across the entire iteration cycle: reviewing draft one for contradictions and untestable requirements, reviewing draft two for the eight issues listed above, drafting the 7-section text from the resolved decisions, and, in the final pass, applying the three fixes identified by an external review of the third draft (see Iteration history, item 4). For three decisions with real trade-offs (transport framing, reservation conflict handling, hard-vs-soft unavailability), I was presented with named options and made the selection myself rather than accepting an AI-proposed default; those are recorded above with the rejected alternatives. The mechanical fixes (thresholds, role scoping, audit logging, assumption ranking, token quota timing, and the final-pass Contract precision edits) were proposed by the AI with reasoning and I reviewed and accepted them without changing the substance.

Self-review notes

- Checked that every Non-Goal in this version still matches something the brief itself lists as out-of-scope for Phase 1 (module decomposition, schema, UI, deployment architecture, exact stack); the one addition in the final pass (hardware-equivalence algorithm) is a scope boundary implied by the Contract's own deferral note, not a new exclusion invented independently.
- Checked that the reservation-conflict Capability and the corresponding Contract behavior describe the same three-step sequence; they do not diverge.
- Checked, after the final pass, that the token-quota timing rule now appears identically in both the Contract and this justifications document; this was the one place an external review found the two documents disagreeing (the decision existed here but had not been carried into the spec itself), and it is corrected.
- Open gap I am aware of and not claiming to have resolved: the "equivalent-or-better substitute" comparison logic itself (how the system decides one node is an adequate substitute for another) is left undefined on purpose; the Contract and Non-Goals now both say so explicitly, so this reads as a stated deferral rather than a silent gap, but the underlying question is still open for a later iteration.
- Open gap: behavior under a large simultaneous submission surge (e.g., a full cohort submitting at once) is flagged in Assumptions as risky but not designed for; if that scenario matters for grading, it needs its own follow-up spec.